

TDDD50 Report

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AI tools used: To fix grammatical and LaTeX syntax

1 Summary

1.1 The main Ideas

The primary focus of this paper is to research the current movement Waste Electrical and Electronic Equipment (WEEE) from developed to developing countries. As a possible solution for WEEE management, the paper presents the regional and global regulations and bans implemented, such as Extended Producer Responsibility (EPR). By adopting bans like EPR, producers are responsible for taking back WEEE products in order to have a smooth recycling system. To achieve even more efficient recycling of WEEE, the paper introduces a new type of EPR called Full Extended Producer Responsibility, which makes manufacturers responsible not only for the WEEE in their own countries but also in the countries to which their WEEE has been exported. Additionally, Due to the hazardous materials like lead, mercury, and accumulators contained in WEEE, its transport results in numerous negative consequences, such as environmental damage and health issues in developing countries.

1.2 Methods

While the methodologies used in the paper are not clearly mentioned, I understood that the methodology chosen for this paper is to select 8 other studies that provide an overview of countries involved in the WEEE movement. In total, there are 32 countries (14 source countries and 18 destination countries) involved in these studies. The paper refers frequently to laws and other existing research papers as well.

1.3 Conclusions

The conclusion outlines the impacts of global and regional regulations on reducing the transportation of Waste Electrical and Electronic Equipment (WEEE) and Used Electrical and Electronic Equipment (UEEE) from developed to developing countries. Although many countries have implemented bans on the import and export of e-waste, illegal transportations continues. The permissions or bans between involved countries can be categorized as banned, permitted, or having no legislation for the export and import of e-waste. It is notable that one key reason behind the decision to ban the import of WEEE in some destination countries like Pakistan and Malaysia is the lack of standard recycling systems and

facilities. Furthermore, to make producers responsible for the entire lifecycle of WEEE products, Extended Producer Responsibility (EPR) is implemented. However, EPR is not implemented in all destination countries. For example, among African countries, only Kenya, South Africa, and Nigeria have implemented EPR. But all source countries have established EPR regulations.

2 Discussion

The article covers a crucial global issue that remains unsolved today and continues to cause various environmental and health problems around the world, especially in developing countries. To address these problems, the paper presents various solutions in terms of global regulations like the Basel Convention and regional regulations and bans, which is a strength of the paper. However, in my opinion, the paper could be more interesting if the authors had conducted some experiments themselves to make their claims and conclusions more reliable. One such experiment could be to choose one of the destination countries as a sample and study the regional impact of WEEE on the lives of people living close to informal recycling sectors, as well as the effects on the environment.

Additionally, the paper explains what the issue is but not specifically why the issue exists. To answer the question of why, the paper could include other crucial factors, such as the economic reasons behind the movement of WEEE from developed countries to developing countries. The study done by Abalansa [1] explains that many developing countries are paid to receive e-waste from developed countries. To address this issue, a good approach could be to track and ban such payments. Instead, the funds could be redirected to support proper e-waste recycling within developed countries.

Even though the idea of EPR, among other bans, is a good solution for e-waste management, it is unrealistic to expect that EPR will be fully implemented in all involved countries because the regulations and facilities vary significantly from one country to another. However, the paper concludes that EPR is not successfully implemented in all destination countries, with only 33% of destination countries in Africa and 44% of destination countries in Asia having established EPR principles. The paper also mentions that the progress of implementing recycling systems in the involved countries, especially in the destination countries, has

resulted in either formal or informal sectors. There is evidence mentioned in the paper that explains some exporters and importers circumvent regulations like the Basel Convention by registering WEEE as Used Electrical and Electronic Equipment (UEEE). These WEEE often end up in the informal sectors in the destination countries.

The conclusion section focuses on explaining the impacts of implementing regulations like the Basel Convention and bans such as EPRR on the movement of WEEE from developed to developing countries. The data presented (on other sections) mostly support the conclusions drawn by the researchers, but the presentation of the data could be improved. In the conclusion section, the researchers show the data in percentage form, while the related data is presented in tables and figures. A better approach to presenting the related data to the data in the conclusion would be to use charts or graphs that include the percentage values as well.

In conclusion, after discussing the paper in the seminar group, the paper covers the aims made by the authors and gives readers an overall picture of the movement of WEEE (Waste Electrical and Electronic Equipment) from source to destination countries. However, the movement of the WEEE is not done in a completely legal way. One of the main topics during the discussion in the group was the continuing illegal WEEE transport despite the established regulations and bans in the involved countries. This discussion ended with the assumption that there are other factors, such as corruption, cultural, and economic factors, causing the illegal movement beyond those covered by the paper. Because there are too many links to other researchers and related works in this article, it is recommended to read those study sources as well to get a clear picture of how e-waste is transported around the world.

3 Related work

Research done by Ilankoon [2] shows that the generation of e-waste is growing rapidly, at a rate of 3-5% per year globally (6.1 kg per person) over the past few years. This trend has consistently been followed by damaging consequences for both nature and human health. Several studies conducted on people working in conventional e-waste management, including children under six years old, show that toxic substances have been found in their blood, hair, and urine. While of the environmental and health damaging consequences caused by WEEE is discussed in a overall manner in my papre, this article dives in a more details to the health damages caused by hazardous substances. To make recycling systems such as take-and-reuse more efficient and widely applied, different e-waste management strategies like collective and clearinghouse approaches have been implemented. The main idea is to make the Extended Producer Responsibility (EPR) principle more suitable and adaptable based on the economic factors in each local region (EPR and e-waste management are related in both papers).

The research done by Shittu [3] explains the definition of WEEE in greater depth by categorizing the materials present in WEEE (Waste Electrical and Electronic Equipment) into different categories depending on the design and functionality of the device. This categorization is meant to help households easily sort e-waste by its end-of-life phase. The paper also explores newer methods and strategies to manage WEEE effectively. These methods are based on regulatory levels and environmental impacts and can vary from one region to another around the world. The newer recycling approaches presented in this paper are strongly connected to the 'take-back' strategy within the context of EPR in my paper. The paper predicts that the production of e-waste will continue to grow globally in the future. It also warns about the introduction of newer products that can make the recycling process more challenging and complex. By providing easier internet connectivity around the world, especially in developing countries in Asia and Africa, the usage of electronic devices such as personal phones, tablets, and televisions will increase as well. In my opinion, more bans like EPR and Full Extended Producer Responsibility (discussed in my paper) will be introduced to make recycling systems more efficient in the future.

4 biography

References

- [1] Samuel Abalansa et al. “Electronic Waste, an Environmental Problem Exported to Developing Countries: The GOOD, the BAD and the UGLY”. In: *Sustainability* (2021). DOI: [10.3390/su13095302](https://doi.org/10.3390/su13095302). URL: <https://www.mdpi.com/2071-1050/13/9/5302>.
- [2] I. M. S. K. Ilankoon et al. “E-waste in the international context – A review of trade flows, regulations, hazards, waste management strategies and technologies for value recovery”. In: *Waste Management* (2018). DOI: [10.1016/j.wasman.2018.10.018](https://doi.org/10.1016/j.wasman.2018.10.018). URL: <https://doi.org/10.1016/j.wasman.2018.10.018>.
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